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fwhFoam: A Ffowcs Williams--Hawkings solver with on-surface acoustic source localization for OpenFOAM --Manuscript Draft--

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fwhFoam: A Ffowcs Williams–Hawkings solver with on-surface acoustic source localization for OpenFOAM

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Abstract

fwhFoam is an open-source aeroacoustic post-processing library for OpenFOAM. It predicts far-field sound with Farassat’s formulation 1A of the Ffowcs Williams–Hawkings equation for permeable and impermeable integration surfaces through one code path, with a uniform mean flow handled in closed form. From the same surface data it also computes the surface sound-power density σ of the diffraction-filter theory of Delfs & Ruck, which maps where the radiated sound originates on the surface, subject to the power identity $\oint \sigma dS = P_{\text{far-field}}$. Far-field predictions match analytic monopole, dipole and convected-monopole solutions to $\sim 0.1\%$, and the σ identity is verified to a few tenths of a percent on analytic sources.

Keywords: Aeroacoustics, Ffowcs Williams–Hawkings, Acoustic source localization, Surface sound-power density, OpenFOAM

Required Metadata

Current code version

1. Motivation and significance

Aerodynamically generated sound is a central concern in the design of aircraft, rotors, wind turbines, ground vehicles and cooling fans. Direct computation of the acoustic field together with the flow is prohibitively expensive for most engineering configurations, because the acoustic fluctuations are orders of magnitude smaller than the hydrodynamic ones and the far field lies many wavelengths from the source. The established remedy is an *acoustic analogy*: the near-field unsteady flow is computed by scale-resolving CFD, and an integral formulation propagates the sound to far-field observers at negligible additional cost. The Ffowcs Williams–Hawkings (FW-H) equation [1] is the

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Nr.	Code metadata description	Value
C1	Current code version	v1.0.0
C2	Permanent GitHub link to code/repository	https://github.com/Sparsh-Sharma/fwhFoam
C3	Legal Code License	GPL-3.0-or-later
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	C++ (C++14), Python 3, OpenFOAM (<code>wmake</code>)
C6	Compilation requirements, operating environments & dependencies	OpenFOAM v2306; C++14 compiler; Python ≥ 3.9 with NumPy and SciPy
C7	Link to developer documentation/manual	README.md, <code>doc/theory.md</code> , <code>doc/fwh-data-format.md</code> in the repository
C8	Support email for questions	sparsh.sharma@dlr.de

Table 1: Code metadata (mandatory)

most general such analogy, and in its permeable (porous-surface) form [2, 3] accounts for volume (quadrupole) sources by enclosing them within an off-body integration surface. Farassat’s formulation 1A [4] is its standard time-domain realization.

Mature open-source FW-H implementations for OpenFOAM already exist; the most established is `libAcoustics` [6], providing Curle, Farassat 1A (permeable and impermeable) and a wind-tunnel formulation. `fwhFoam` does not aim to replace such tools on far-field prediction, and we claim no novelty for the FW-H formulation, which is standard. It makes a compact, verifiable FW-H capability openly available with a reproducible verification and convergence suite. In addition, it couples the far-field prediction with on-surface source localization: the same permeable-surface data used for the FW-H integral are processed into a surface sound-power density $\sigma(\mathbf{x})$ that identifies which parts of the surface radiate to the far field. This quantity derives from the diffraction-filter theory of Delfs & Ruck [7] and locates the acoustic sources on a surface — information the far-field signal does not carry — which existing OpenFOAM FW-H tools do not provide. The identity $\oint \sigma dS = P_{\text{far-field}}$ relates the two and serves as an internal consistency check. The user drives `fwhFoam` either through a runtime *function object* that samples surface data during an OpenFOAM run and writes observer signals, or offline via the `fwhSolve` utility, which recomputes observer signals (and, with

the Python tools, the σ map) from stored surface data without re-running the CFD.

2. Software description

2.1. Software architecture

`fwhFoam` is built around a mesh-independent formulation engine (`fwhFormulation1A`) shared by two front-ends: a compiled OpenFOAM library `libfwhFoam` exposing the runtime function object `fwh`, and the stand-alone utility `fwhSolve`. The engine operates on raw face-centre data (position, unit normal, area, and per time-step samples of gauge pressure, density and fluid velocity), so identical numerics serve the patch-based (impermeable) and sampled-surface (permeable) paths and the runtime and offline front-ends. A Python package `pyfwh` provides surface-data I/O, analytic reference fields, spectra, integration-surface generators and — through its `localization` module — the surface sound-power density σ ; the driver `sigma_localize` produces σ maps from surface-data files.

2.2. Far-field formulation

The acoustic pressure is split into thickness (T) and loading (L) contributions. For a static surface S with outward unit normal $\hat{n}$ in a uniform mean flow of Mach vector $\mathbf{M} = -\mathbf{U}_0/c_0$,

$$4\pi p'_T = \int_S \left[\frac{\rho_0 \dot{U}_n}{r(1-M_r)^2} + \frac{\rho_0 U_n c_0 (M_r - M^2)}{r^2(1-M_r)^3} \right]_{\text{ret}} dS, \quad (1)$$

$$4\pi p'_L = \int_S \left[\frac{\dot{L}_r}{c_0 r(1-M_r)^2} + \frac{L_r - L_M}{r^2(1-M_r)^2} + \frac{L_r c_0 (M_r - M^2)}{r^2(1-M_r)^3} \right]_{\text{ret}} dS, \quad (2)$$

with permeable-surface source vectors $U_i = v_i + (\rho/\rho_0)(u_i - v_i)$ and $L_i = p'\hat{n}_i + \rho u_i(u_n - v_n)$. For a solid wall $u_n = v_n$ and the classical impermeable form is recovered, so no separate code path is needed. Rather than solving the implicit retarded-time equation, each source time-level is propagated forward to the observer at $t = \tau + T$; the emission-to-reception delay T under a uniform flow follows from the closed-form Garrick-triangle relation $(c_0^2 - U_0^2)T^2 + 2(\mathbf{d} \cdot \mathbf{U}_0)T - |\mathbf{d}|^2 = 0$, requiring only $|\mathbf{U}_0| < c_0$. Each contribution is deposited onto a uniform observer-time grid by exact piecewise-linear resampling; the scheme is single-pass and parallelises by summing per-rank partial signals.

2.3. On-surface source localization

Working per angular frequency $\omega = c_0 k$ with the complex surface spectra $\phi = \hat{p}$ and $\psi = \partial \hat{p} / \partial n = -i\omega \rho_0 \hat{v}_n$, the radiating field has far-field directivity $F(\hat{\mathbf{x}}) = \frac{1}{4\pi} \oint [ik(\hat{\mathbf{x}} \cdot \hat{\mathbf{n}})\phi - \psi] e^{ik\hat{\mathbf{x}} \cdot \boldsymbol{\xi}} dS$ and power $P = \frac{1}{2\rho_0 c_0} \oint |F|^2 d\Omega$. Performing the angular integration in closed form yields the exact surface sound-power density

$$\sigma(\boldsymbol{\xi}) = \frac{1}{32\pi^2 \rho_0 c_0} \operatorname{Re} \oint [k^2(\hat{\mathbf{n}} \mathbf{W} \hat{\mathbf{n}}') \phi \phi'^* - ik(\hat{\mathbf{n}} \mathbf{V}) \phi \psi'^* + ik(\hat{\mathbf{n}}' \mathbf{V}) \psi \phi'^* + J_0 \psi \psi'^*] dS', \quad (3)$$

with $W_{ij} = 4\pi[\delta_{ij}j_1(z)/z - \hat{d}_i \hat{d}_j j_2(z)]$, $V_i = 4\pi i j_1(z) \hat{d}_i$, $J_0 = 4\pi j_0(z)$, $\mathbf{d} = \boldsymbol{\xi} - \boldsymbol{\xi}'$, $z = k|\mathbf{d}|$. By construction $\oint \sigma dS = P$: the localization map and the far-field power are two views of the same data and must agree. On a rigid wall ($\psi = 0$) only the pressure block survives, recovering the classical rigid-surface density [7]. The density is signed, reflecting that radiated power is a two-point quantity.

3. Illustrative examples

3.1. Far-field verification and convergence

Exact acoustic fields placed on a geodesic icosphere enclosing the source are propagated and compared with the analytic far field. The monopole (thickness term), dipole (loading term) and a convected monopole at $M = 0.3$ (Garrrick delay and convective terms) all agree to a relative L_2 error of order 10^{-3} over the radiating directions. Refining the surface resolution and the time step (Fig. 1) yields second-order convergence in both (observed orders 2.6 and 2.3), consistent with midpoint-rule surface quadrature and central-difference time derivatives.

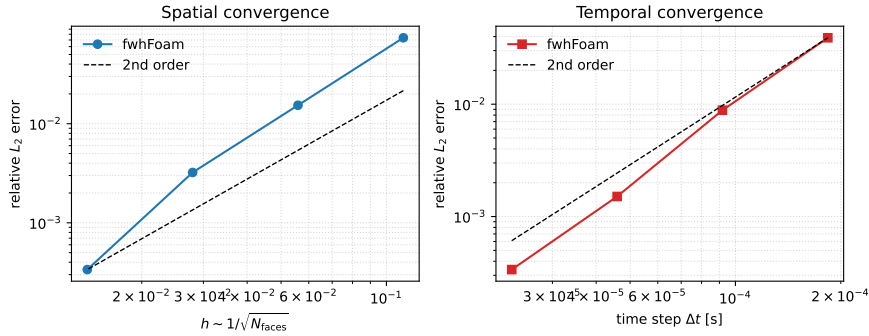


Figure 1: Spatial (left) and temporal (right) convergence of the relative L_2 error for the analytic monopole; dashed lines are second-order slopes.

3.2. Verification of the localization density

The power identity underlying σ is verified independently: for exact monopole and dipole fields on a 1280-face icosphere, the analytic radiated power, the far-field power and $\oint \sigma dS$ agree to 2.9×10^{-3} , and σ reproduces the correct spatial structure — uniform for the isotropic monopole, and vanishing at the poles while peaking at the equator for the dipole ($\cos^2$ pattern).

3.3. Aeolian tone and source map of a circular cylinder

Laminar flow past a 2D cylinder at $Re = 100$ (incompressible `pimpleFoam`) sheds a von Kármán street, producing a lift-dipole tone. The runtime `fwh` object on the cylinder wall recovers the tone at the shedding Strouhal number $St = 0.166$ (Fig. 2), matched against the lift-coefficient spectrum, with the expected dipole directivity. Sampling instead on a permeable control surface of radius $8D$ and applying (3) at the tone yields the source map of Fig. 3: the density concentrates on the downstream sector where the unsteady wake crosses the surface, forms a signed lobe pair, and integrates to the far-field power to machine precision — the built-in identity holding on genuine CFD data.

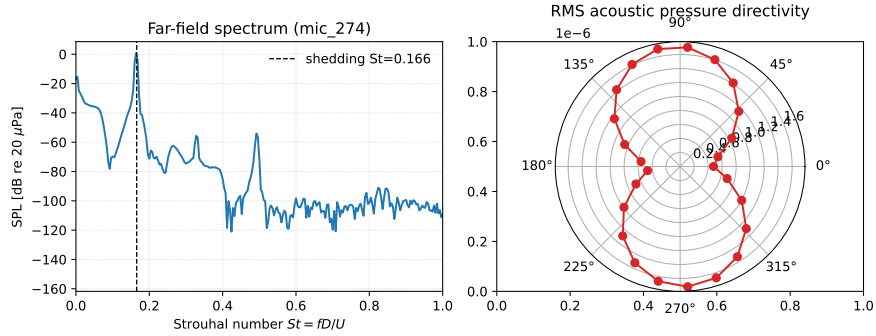


Figure 2: Cylinder Aeolian tone at $Re = 100$: (left) far-field spectrum at the cross-flow observer, peaking at the shedding Strouhal number; (right) RMS acoustic-pressure directivity (lift-dipole lobes).

4. Impact

As a far-field solver, `fwhFoam` provides a documented and verified FW-H capability that complements existing libraries; the offline `fwhSolve` utility separates the CFD from the acoustic post-processing, so observer arrays, ambient conditions and mean-flow speeds can be changed without re-running the flow solver. The surface sound-power density σ additionally turns an FW-H integration surface, normally used only for a far-field signal, into a

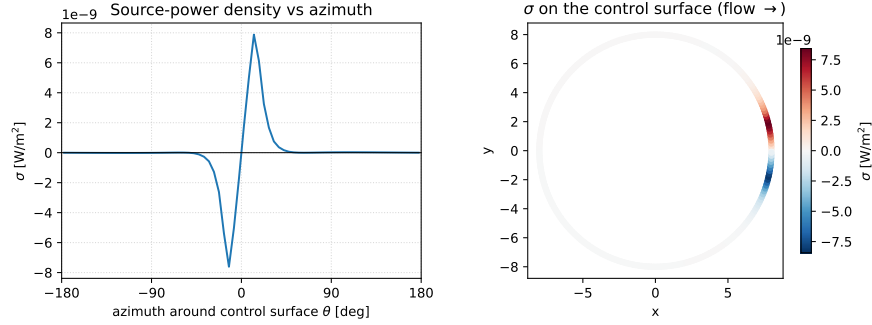


Figure 3: Source-power density σ on the permeable control surface ($r = 8D$) at the shedding tone: (left) azimuthal profile; (right) σ over the surface (mean flow left to right). The signed density localizes the radiating region; $\oint \sigma dS = P$ to machine precision.

diagnostic of which surface regions radiate. This is relevant to noise-reduction studies, where the location of the sources on an airframe, edge or porous treatment matters and is not available from a far-field spectrum alone [7]. Because σ uses the same data as the FW-H integral and satisfies the power identity, it adds this diagnostic without additional simulation and with an internal check. The verification and convergence suites are reproducible, and the mesh-independent engine is a basis for moving-surface and frequency-domain extensions.

5. Conclusions

`fwhFoam` is a Ffowcs Williams–Hawkings solver for OpenFOAM that supports permeable and impermeable surfaces and a uniform mean flow, and that computes the surface sound-power density σ for on-surface source localization from the same data as the far-field integral, subject to a power identity used as an internal check. Far-field predictions are verified against analytic solutions to about 0.1% with second-order convergence, and the localization density is verified through its power identity to a few tenths of a percent. Planned extensions include moving and rotating integration surfaces, the diffraction-filter projection for rigid-wall data, and a frequency-domain formulation.

Acknowledgements

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: